

NASA Digital Engineering Windchill Migration Services

Multi-Center Windchill (MCWC) Migration and Consolidation
Initiative

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Executive Overview

Overview – NASA Digital Engineering PLM Systems Consolidation

NASA Digital Engineering Program is responsible for improving how the Agency Engineering Domain operates over the entire NASA lifecycle by effectively managing complexity, reducing cost and schedule, and improving product integrity via the integration of processes, digital tools, and techniques along with seamless flow of information throughout the engineering system development lifecycle (concept development, design, testing and validation, manufacturing and operations).

To fulfill these responsibilities, the consolidation of like assets is needed to not only reduce operational overhead costs, but to increase “findability” and reuse of engineering designs and solutions, increased consistency of data management practices, and increase integration of tools and information repositories within an enterprise setting to only name a few.

As NASA’s digital engineering data thread evolves, with associated tool/information repository integrations, having a core product lifecycle management (PLM) system which consolidates the bulk of NASA Engineering and Project information is important and will act as the cornerstone of the engineering data thread going forward.

To address this need, NASA’s HQ Office of Chief Engineer (HQ OCE) seeks a highly qualified service provider to plan, design, migrate, validate, and migrate multiple NASA Windchill v13 instances located in on-prem data centers into an existing cloud-based Windchill v13 instance from here forward will be referred to as the Multi-Center Windchill Cloud (MCWC) in AWS GovCloud.

Section I. Purpose and Intent

The objective of this acquisition is to migrate existing Center Windchill PLM systems into the Agency Multi-Center Windchill Cloud platform. The goal would have a base contract to provide migration services to one NASA Center with options to perform additional migrations. Cost effective the options may be turned on before the completion of the base contract if it is cost beneficial to NASA.

NASA intends to procure a capable service provider to deliver a comprehensive migration effort that merges multiple Windchill v13 instances into the pre-existing MCWC. The engagement will require data gathering on exiting systems (which will be referred to as Source Instances), migration planning with pre-migration testing, execution of the migration itself, and a post-migration wrap-up focused on issue resolution, and knowledge transfer to the MCWC operational team. A central objective is to ensure that every Source Instance’s data migrates completely and per expectations of the Source Instance owners into the MCWC. The successful bidder will demonstrate superior knowledge and experience in enterprise-level Windchill operations, cloud-based migrations, and a proven ability to

execute complex data merges with minimal disruption to end users which is important as these Source Instances support large, operational NASA programs.

Proposers should anticipate integrating source data into the MCWC while preserving data fidelity, access controls, licensing compliance, and organizational governance.

Section II. Background

NASA maintains several Windchill v13 deployments in on-prem data centers in different geographical locations. The MCWC (the current JSC Windchill system and will be the target system of which there are no current planned changes in terms of existing Orgs, workflows, integrations that exist prior to other centers migrating in) exists as a centralized Windchill v13 environment in the AWS GovCloud with operational Dev/Test/Production environments and an established architecture and security plan. The migration effort will merge initially **(qty 3)** on-prem Windchill v13 Source Instances into the MCWC, with every Windchill “Org” in the Source Instances migrating into the MCWC Org framework; however, based on service provider performance additional center migrations may be desired. Proposers must demonstrate (provide relevant examples via names of past clients, types of migration services, and complexity) extensive Windchill migration experience, capabilities migrating into – and operating within – the AWS Cloud environment, and a track record of working within government-regulated environments. The governance framework for MCWC – encompassing data governance, security controls, access management, and auditability – will guide the migration activities. Key stakeholders include the NASA HQ OCE Digital Engineering Program Manager, Source Instance Owners and sys admins, and supporting NASA Agency IT personnel.

Note: Quotes should reflect an ability to operate within the MCWC as-is, including honoring the existing Org constructs and ensuring clean Org information mapping with conflict-resolution procedures when merging multiple source Orgs into MCWC Org structures.

Section III. Scope of Work

The migration and consolidation of each PLM instance will likely consist of the following general phases:

- **Data Gathering** – The contractor will assess all on-prem Windchill v13 instances, catalog data domains, customizations, integrations, document data schemas, workflows, security roles, access controls, Org structures, and get agreement with the Source Instance owner what will need to be migrated vs what will be left behind and expected organization of information and functionality once the migration is complete. A critical deliverable is an explicit Org mapping plan that details how each source Org will map to an MCWC Org, along with recommendations for data to migrate versus data to archive or leave behind.

- **Migration Planning and Testing** – Each migration approach will be documented in a Migration Plan that describes sequencing, pre-migration checks, rollback criteria, and cutover windows, all designed to integrate with MCWC governance and will be agreed to by both the MCWC sys admin and the Source Instance owner/sys admin. A testing strategy will be defined, covering functional, performance, security, and user acceptance testing, with explicit acceptance criteria including Org mapping validation. A Migration Validation Plan will be prepared to prove data fidelity, object references, lifecycles, permissions, and Org integrity within MCWC before production cutover. A Communication Plan will be developed to inform stakeholders of changes, risks, and sign-off points, aligned with MCWC governance.
- **Executing Migration** – The actual migration from on-prem Windchill v13 instances to MCWC in AWS GovCloud will be executed in accordance with the NASA-approved Migration Plan of the service provider which will include an expected risks and associated mitigation plans. The migration will utilize standard Windchill migration methods and approved automation scripts, with no expectation for new bespoke tool development as part of these migrations or which needs to be maintained post-migration. Progress will be monitored, issues managed, and adjustments made to stay on plan, with migration activities in accordance with governance and security requirements for data in transit within the NASA cybersecurity environment.
- **Post-Migration Wrap-up** – After cutover of each Source Instance to MCWC, remaining issues surfaced during testing and migration will be resolved, including any Org-merge conflicts to be resolved by the service provider. Post-migration incident response procedures, and knowledge transfer sessions will be delivered to both the MCWC sys admin and Source Instance owner/sys admin. A final Source Instance Migration Closeout Report will summarize lessons learned.

Section IV. Technical and Functional Requirements

Migration Expertise, Services, and Support

The Windchill PLM migrations needed will involve the movement of large volumes of information, files, and meta data associated with over two decades of NASA engineering, with current/active Program and project information from existing Source Instance Windchill environments to the existing MCWC system residing in the NASA partition of AWS' Gov Cloud environment.

Typical functions expected from this services procurement include but are not limited to:

- Superior knowledge and experience in the operation of enterprise-level Windchill systems, and the migration of data/information from one Windchill system to another with no unexpected data loss.

- Experience in working within cloud-based environments and moving data/information into the cloud environment from outside the cloud from on-premise data sources.
- Ability to fully understand the structure and organization of the source Windchill system and the understanding of how to map and migrate into the destination Windchill system, communicate the plan to the owners of the source system and the challenges migrating from one data schema to another, negotiate any changes to how information is organized in the destination system, work with the owners of the destination system to communicate the plan, any changes needed on the destination system, and ability to fully and successfully execute of the agreed to plan.
- Ability to successfully execute the migration and post-migration activities with tools sufficient to execute the migration – the expectation being there will be no need for the development of any custom tools to perform the migrations beyond minimal automation scripts with no expectation for new bespoke tool development as part of these migrations or which needs to be maintained post-migration.

Based upon service provider performance, other Source Instance migration associated with additional TBD centers will be considered; therefor provide estimate for additional migrations which can be enacted by NASA as a contract option which would be in addition to the base period of performance of the contract.

Implementation

The current NASA MCWC is set up such that each Center, major Organization, or Program are considered their own “Org” in the Windchill structure. Therefore, Source Instances may have a combination of Center Organizations and Programs which will migrate such that they are all at the same level, per figure 1, but will ultimately be at the discretion of the Source Instance owner/sys admin. Note, if there are numerous small projects or orgs, the owner of the Source Instance may opt to have them grouped in a single Org in the destination MCWC system.

Figure 1: General MCWC structure showing how Source Instance PLM content can migrate into general Center Orgs within the MCWC or Programs or large orgs can migrate into their own MCWC Orgs.

Section V. Data Migration and Configuration Requirements

- **Source Data Inventory** - The contractor shall develop a comprehensive inventory of each Source Instance, including but not limited to:
 - Object counts by type (e.g., WTPart, WTDocument, EPMDocument)
 - Customizations (e.g., workflows, subtypes, attributes)
 - Data volumes (vault sizes, attachment counts)
 - Known data quality issues (e.g., duplicates, orphans, broken references)
 - Integration points (e.g., CAD, ERP, external repositories)

This inventory must be validated with the Source Instance owner and used to inform the Migration Plan and Org Mapping Strategy.

- **Migration Methodology** - Each Source Instance migration shall follow a multi-phase, incremental approach aligned with the MCWC Reference Architecture. The methodology must include:
 - Pre-migration readiness assessments
 - Sequenced migration phases (e.g., pilot, full cutover)
 - Defined cutover windows and rollback procedures
 - Risk mitigation strategies for high-volume or high-complexity data sets

The contractor shall ensure minimal disruption to ongoing operations and provide clear communication protocols for each phase.

- **Windchill Version and Platform** – The MCWC must continue to operate on Windchill v13.x, with support for the required modules currently in the Production environment. All source Windchill v13 instances reside on-prem; the MCWC resides in AWS GovCloud and serves as the target environment. The proposer must explicitly reference the existing MCWC architecture and demonstrate how the migration will conform to security controls and MCWC data structures. Any proposed deviations must go through formal MCWC and Source Instance owners and sys admins.
- **Data Management and Migration** – The plan must address data gathering, normalization, and mapping to the MCWC schema, ensuring preservation of metadata, lifecycles, histories, attachments, and audit trails to the expectations of the Source Instance owner. Data cleansing, deduplication, and normalization should be performed as needed, with the agreed-to Org-level mapping rules.
- **Interoperability and Integrations** – The migrations must maintain functionality of existing PLM integrations with other data repositories, CAD pipelines, and external data sources, with API compatibility and integration testing to ensure seamless operation within MCWC's consolidated Org structure.
- **Data Transformation and Mapping** - The contractor shall define and document:
 - Source-to-target mapping rules for all object types
 - Org mapping strategy (e.g., one-to-one, consolidated, or hybrid)
 - Handling of legacy data (e.g., version depth, obsolete objects)
 - Conflict resolution procedures for naming, permissions, and lifecycle mismatches
 - **Documentation** – All deliverables must include updated architecture diagrams, data migration reports, and test results reflecting the merged MCWC state.

All mapping decisions must be reviewed and approved by the Source Instance owner and MCWC governance team.

Validation and Verification – Post-migration validation shall include:

- Reconciliation reports comparing source and target object counts, metadata, and relationships
- Verification of lifecycle states, access controls, and audit trails
- Functional testing of workflows and integrations
- User acceptance testing (UAT) with Source Instance stakeholders

Acceptance criteria must be clearly defined and signed off prior to final cutover.

- **Configuration Management** – The contractor shall ensure that:

- Source Instance configurations (roles, ACLs, workflows) are preserved or adapted to MCWC standards
- Customizations are reviewed for compatibility and necessity
- Any deprecated or unsupported configurations are documented and retired in coordination with the Source Instance owner
- Configuration changes must be traceable and compliant with MCWC change control procedures.

Section VI. Architecture, Security, and Compliance

- **Reference Architecture** – The existing MCWC architecture and current operational environments will guide the migration; updates to architecture will occur only through formal MCWC review and documented change requests.
- **Security Controls** – proposer activities must comply with NASA identity management, authentication, authorization, encryption, logging, monitoring, and incident response must align with the MCWC plan and GovCloud requirements.
- **Compliance** – Evidence of certifications, assessment results, remediation plans, and ongoing compliance posture relevant to MCWC in GovCloud should be provided or demonstrated upon request.
- **Data Residency and Sovereignty** – A clear policy on data location within GovCloud, GovCloud - on-prem data flows, and related controls must be articulated and adhered to within MCWC governance.

Section VII. Deliverables, Milestones, and Acceptance

- **Deliverables** – Source Instance Data Gathering Report including an Org Information Mapping Plan; Recommended Final MCWC System Sizing which will provide any recommendation to upsizing/adding to existing MCWC servers as part of pre-migration approvals; a Source Instance Migration Plan aligned with MCWC governance; Source Instance Migration Validation Reports covering data fidelity and associated metrics which will be used to demonstrate a lossless data transfer, Org integrity, and permissions; a Source Instance Cutover Plan with Rollback Procedures; Source Instance Migration Knowledge Transfer sessions as needed; and a Source Instance Migration Closeout Report with migration lessons learned.
- **Milestones** – The project will follow a cadence beginning with a contract ATP kickoff, followed by the following for each Source Instance migration: Data Gathering and Org Mapping completion, Design and Plan approval within MCWC governance, Migration Validation sign-off, Phase 1 pre-migration into MCWC (pilot test), Phase 2 migration (full migration and cutover) into MCWC, stabilization and handover, and a final closeout for each Source Instance migration. The Schedule is aligned with ATP

through 1 year, with specific weeks defined in the project plan and subject to governance approvals.

- **Periodic Reviews** – Weekly status reviews including any risks or issues with MCWC and Source Instance stakeholders. At the completion of each Source Instance migration, there will be a review of all mentioned deliverables and required buy-off by Source Instance owner that migration has occurred to their expectations.
- **Acceptance Criteria** – All critical workflows must be migrated and validated within MCWC; data migration fidelity and Org information mapping must be confirmed via predefined reconciliation metrics; the cutover must be completed with no unmitigated critical incidents; Org access controls must be implemented and compliant with stated requirements. All these must have buy-off by Source Instance owners.
- **Warranty and Support** – A minimum no-cost 90-day post-migration warranty will be provided for each Center migration.

Section VIII. Performance Metrics

To ensure a high-quality migration with minimal disruption to NASA operations, the following performance metrics shall be used to evaluate contractor performance throughout the engagement:

Data Fidelity

- $\geq 99.9\%$ of migrated objects must retain original metadata, version history, and relationships.
- $\leq 0.1\%$ data loss or corruption rate, verified through post-migration reconciliation reports.

Migration Accuracy

- 100% of Org mappings must match the approved Org Information Mapping Plan.
- All critical workflows and permissions must be validated and functional post-migration.

Downtime Tolerance

- Planned cutover windows must not exceed the agreed downtime tolerance per Source Instance.
- Unplanned downtime must not exceed 4 hours per Source Instance during migration.

Issue Resolution

- $\geq 95\%$ of post-migration issues, non-critical must be resolved within 10 business days of cutover.

- Non-Critical issues would be but not limited to any instance where the user can still work without having to redo their work once issue has been corrected
 - Indexing errors less than 20%
- Critical issues must be triaged and addressed within 24 hours after identification.
 - Critical issues would be but not limited to any system down, user's down, user's are able to use it but might not be able to be fix it without redoing the work already done, issues with user's data that doesn't get processed through Center's workflow and Security Labels/Controls not working

Schedule Adherence

- $\geq 90\%$ of milestones must be met within ± 5 business days of the baseline schedule.
- Any delays must be communicated to NASA COR with mitigation plans within 48 hours.

Stakeholder Satisfaction

- $\geq 90\%$ satisfaction rating from Source Instance owners based on post-migration surveys.
- All migrations must receive formal sign-off from Source Instance owners based upon the above mentioned performance metrics .

Security Compliance

- 100% compliance with MCWC and NASA cybersecurity controls, verified by audit logs and security scans.

Section IX. Program Management, Staffing, and Location of Work

- **Program Management** – The project will follow a MCWC-aligned governance model, including defined roles, responsibilities, escalation paths, change control, risk management, and a formal communications plan.
- **Staffing Requirements** – The engagement likely requires Windchill architects, cloud engineers, data migration specialists, security/compliance leads, QA/testers, and a program manager with experience in Windchill migrations and GovCloud deployments.
- **Location of Work** – Work will be conducted in a hybrid fashion (on-site, remote, or mixed) with security controls and data handling aligned to GovCloud requirements and MCWC governance.
- **Travel** – Travel will be as-needed and pre-approved, with cost allowances as appropriate.

Section X. Evaluation and Award

- **Basis for Award** – The award will be based on best value considering technical merit, past performance, feasibility, staffing qualifications, and best value to the government.
- **Evaluation Factors and Weights** – Proposed Technical Approach and Migration Plan (shall include the specific migration tool(s) which will be used and how they will address the known challenges of migrating well-established med-to-large Windchill systems based on past experience); Past Performance and Experience (Windchill migrations and GovCloud); Key Personnel Windchill Qualifications; Non-Windchill Technical Skill Competence (Security, Architecture, Cloud, Org Governance); Price will be equally weighted.
- **Evaluation Process** – Quotes will undergo an initial responsiveness screen, followed by evaluation by a government-style panel. References and demonstrations may be requested, and final discussions will be conducted with top-scoring bidders.
- **Note** – Quotes will be evaluated on the ability to work within the MCWC as-is, with a clear, auditable path for Source Instance consolidation and data merging.

Section XI. Submittal Instructions

- **Quote Format** – The Technical Quote should describe the approach, architecture alignment with MCWC, schedule, risks, and mitigations. The Management Proposal should include the staffing plan and resumes. Subcontractors must be disclosed, and key personnel must meet minimum qualifications (ability to qualify for a NASA security badge and achieve a Level of Confidence (LOC) 50 clearance) and provide the relevant migration experiences [including experience with the proposed migration tools] of said key individuals. Past Performance should include relevant Windchill migration and GovCloud deployment references/ examples with previous customer names and related migration summary details.

The Price quote should present detailed pricing by deliverable, including labor, travel, licenses, and support costs. Appendices should contain required forms and supplemental information. Professional Services include the activities outlined in sections for proposed migration expertise and services. Provide the total professional services costs for the migration efforts and also broken down per source Center such that they can be structured as contract options.

Identify and itemize the costs for any third-party services included in the quote. The hourly costs must be identified by project role as well as the time required for each task. Note: It is important that the total cost for each Source Instance migration be clearly identifiable and broken down as each Source Instance owner has their own

funding and will make the determination to execute the migration per this contract based upon the information provided for each migration.

- **Page Limits and Formatting** – Adhere to portal requirements and formatting guidelines; ensure readability and consistency. Total quote page count should not exceed **30** pages.
- **Clearances and Certifications** – Include evidence of any required or existing security clearances if applicable and any existing government clearances or NASA identities of personnel who will perform the work if awarded.
- **Clarifications** – Questions should be submitted by the date stated in the procurement schedule; responses will be provided via addenda.

Section XII. Terms and Conditions

- **Contract Type** – The contract is Firm-Fixed-Price (FFP) for migration services and deliverables as defined in the Scope of Work.
- **Invoicing** – Invoicing will be milestone-based, aligned to approved deliverables.
- **Intellectual Property** – Pre-existing IP remains with its owners; MCWC-related deliverables will be governed by negotiation and contract terms.
- **Data Ownership** – NASA data remains owned by NASA, and any industry partner / government contractor IP contained within any of the Source Instance should be handled and protected such that existing IP controls remain in place, with data handling in accordance with policy.
- **Confidentiality** – Standard government non-disclosure agreements are required.
- **Termination** – Provisions for convenience of the government or for cause with defined transition.
- **Authorized Representations** – The bidder warrants accuracy and compliance with all requirements.
- **Security Clauses** – The contract will require compliance with NASA security controls, incident reporting, and attestations.

Appendix A. Technical Information of Source & Destination Windchill Systems

MCWC Destination System (JSC's Windchill) Technical Information

Location: Within NASA's partition of AWS Gov Cloud

Data & Workflow management approach: The MCWC will be built upon the existing JSC Windchill system, with the associated system specs following. While the data within this system will not move, it is being provided such that any proposer knows what is in place now and can plan accordingly. Data vault per each main Org within MCWC to enable clear cloud data storage and compute costs for chargeback business processes. Workflows and customizations should be localized to within the tenant Org's area – any workflow or customization which is required to be at the instance level and affect more than one Org within the MCWC must be agreed to by the MCWC owner.

Current Data Storage footprint: ~50 TB

Source Windchill Instance (I.D./Name and Location)

- Source Instance ID / Name: JEDII / NASA's portion of AWS GovCloud
- Data Center / Location: Amazon GovCloud
- OS Version: Windows Server 2019

Windchill Version and Build

- Windchill Version/Build: 13.0.2
- Patch Level / Interim Fixes: 5

Modules and Scope

- Active Windchill Modules (PDMLink, Product Structure, BOM, etc.): WADM (defunct), ProjectLink
- Module-specific configurations: N/A

Database and Infrastructure

- Database Platform (Oracle/SQL Server/Other): Oracle Enterprise Edition
- Database Version: 19.27.0.0.0
- Database Size (TB):
- Application Servers/Stack Details:
- Non-default Schemas or Special Configs:

Data Volumes and Content

- Total Records/Object Counts:
- Attachments Count and Total Size:
- Number of Vaults & Sizes (TB): 4 vaults (1.4TB, 11TB, 64MB, 44GB)

Data Quality and Clean-up Needs

- Known Data Quality Issues (Duplicates, Orphans, etc.): N/A

- Planned Cleansing/Normalization Steps (if any): N/A

Customizations and Extensions

- Description/number of Custom Workflows/Rules/Events/Job Steps: 44 workflows
- Description/number of UI Customizations and CodeSnippets: Custom attribute filtering on document softtype in library contexts
- Number of Custom subtypes and Attributes: 119 objects, 311 attributes

Integrations and Interfaces

- Interfaces and Data Exchange Cadence: LUNA (in progress), SimManager (in progress)
- CAD Integrations and Data Pipelines: Creo, NX (in progress)

Cutover Readiness

- Proposed Cutover Windows and Downtime Tolerance: Cutover dictated by source, downtime limited to weekends (Friday 6pm-Monday 7am) and weekdays after hours (6pm-7am)

Post-Migration Considerations

- Post-Migration Support Requirements:
- Post-Merge Compliance:

Additional Notable Information:

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Source Windchill System Technical Information by Source

This section will briefly provide some contextual information regarding the version of the source Windchill system along with information regarding size and complexity to inform any quotes.

Kennedy Space Center (KSC)

Source Windchill Instance (I.D./Name and Location)

- Source Instance ID / Name:
- Data Center / Location: KDC / Kennedy Space Center, Florida
- Hosting/Virtualization (On-prem, VM type, hypervisor details): On-Prem
- OS Version: Windows Server 2022

Windchill Version and Build

- Windchill Version/Build: 13.0.2.7
- Patch Level / Interim Fixes:

Modules and Scope

- Active Windchill Modules (PDMLink, Product Structure, BOM, etc.):
- Windchill 13.0.2.0 supplier management
 - Creo View - Clients 11.1.0.0-QDN1AJ
 - Windchill 13.0.2.5 REST services
 - Windchill projectlink 13.0.2.0
 - Creo View - Adapters 11.1.0.0-QDMPSw
 - Windchill 13.0.2.5 service pack
 - Windchill pdmlink 13.0.2.0
 - Windchill 13.0.2.0 info*engine
 - Windchill 13.0.2.0 Services
 - Windchill 13.0.2.0 common base
- Module-specific configurations:

Database and Infrastructure

- Database Platform (Oracle/SQL Server/Other): Oracle
- Database Version:
- Database Size (TB):
 - Actual Data: 200,203 MB
 - Actual Index: 111,611 MB
 - Actual Blobs: 37,059 MB
 - WCAUDIT: 40,715 MB
- Application Servers/Stack Details:
- Non-default Schemas or Special Configs:

Data Volumes and Content

- Total Records/Object Counts: (see below)
- Attachments Count and Total Size:
- Number of Vaults & Sizes (TB):

Data Quality and Clean-up Needs

- Known Data Quality Issues (Duplicates, Orphans, etc.):
- Planned Cleansing/Normalization Steps (if any):

Customizations and Extensions

- Description/number of Custom Workflows/Rules/Events/Job Steps:

- Description/number of UI Customizations and CodeSnippets: ~1000 custom methods/extensions of WC, ranging from increased functionality (export of objects to excel) automated collections, Business algorithms for attribute concatenation etc.
- Number of Custom subtypes and Attributes:

Integrations and Interfaces

- Interfaces and Data Exchange Cadence:
- CAD Integrations and Data Pipelines:

Org Structure and Governance

- Source Orgs Count: 17
- Proposed Org Mapping Strategy [Source Org → MCWC Org] (one-to-one or custom): custom, consolidation
Proposed org going forward, 4, EGS, KSC, <TBD>, <TBR>

Cutover Readiness

- Proposed Cutover Windows and Downtime Tolerance: Artemis Processing/operational system

Post-Migration Considerations

- Post-Migration Support Requirements:
- Post-Merge Compliance:

Additional Notes

- Any Known Constraints or Dependencies: Artemis II

Additional Notable Information:

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KSC Workflows

005 - Notification Route (KSC)
 010 - Route for Peer Review (KSC)

Show Default
 Show Default

migrate Notes

Y
 Y

015 - Export Control Routing (KSC)	Show	Default	Y	
020 - Promote In-Work/Completed/Stored (KSC)	Show	Default	Y	
025 - Move to Library (NE)	Show	Default	Y	
Approval Process	Show	Default	N	OOTB
Change Activity Workflow	Show	Change Management	N	OOTB
Change Notice Workflow	Show	Change Management	N	OOTB
Change Request Workflow	Show	Change Management	N	
Copy of KSC Signature Required Change Activity	Show	Change Management	N	
Core Translation Process	Show	Default	N	OOTB
Design Review Workflow	Show	Change Management	N	OOTB
EMM Assessment Approval	Show	Default	N	
Empty Workflow	Show	Administration	N	
ESC Engineering Change Request Workflow	Show	Change Management	N	
Import Translation Process	Show	Default	N	OOTB
KPLSS Engineering Change CT WF - Flexible	Show	Change Management	Y	
KSC 90% Peer Review Workflow	Show	Change Management	Y	
KSC A-Release CN Wflow	Show	Change Management	N	
KSC Cancellation CN Wflow	Show	Change Management	N	
KSC Change Request Workflow	Show	Change Management	Y	
KSC Described Change Activity	Show	Change Management	Y	
KSC Documentation Release CN Wflow	Show	Change Management	N	
KSC Drafting Check	Show	Engineering	Y	
KSC Engineering Change CN Sustaining WF	Show	Change Management	Y	
KSC Engineering Change CN WF	Show	Change Management	Y	
KSC Engineering Change CT WF	Show	Change Management	Y	
KSC Engineering Change CT WF - DDTE (SMS/GIS)	Show	Change Management	N	
KSC Engineering Change CT WF - Document	Show	Change Management	N	
KSC Engineering Change CT WF - Flexible	Show	Change Management	Y	
KSC Engineering Change CT WF - SI KPLSS	Show	Change Management	Y	
KSC Engineering Change CT WF - Sustaining	Show	Change Management	Y	
KSC Engineering Change CT WF - Sustaining (SMS/GIS)	Show	Change Management	N	
KSC Engineering Upgrade Test CT WF	Show	Change Management	N	
KSC EO CT WF	Show	Change Management	Y	
KSC EO Workflow (Variance)	Hide	Change Management	Y	
KSC General CN Wflow	Show	Change Management	N	
KSC Generic Described CN Wflow	Show	Change Management	N	
KSC Inactive CN Wflow	Show	Change Management	N	
KSC KDDMS IPT CT WF - Flexible Testing	Show	Change Management	N	
KSC NE Container Request	Show	Change Management	Y	
KSC Peer Review Workflow	Show	Default	Y	
KSC PMN Certification Workflow CT WF (KTI-2713C rev C)	Show	Change Management	Y	
KSC Process Action Item WF	Show	Change Management	Y	
KSC Procurement Release CN Wflow	Show	Change Management	N	

KSC Program Model Number Request WF	Show	Change Management	Y
KSC PVS CN CT Workflow	Show	Change Management	Y
KSC PVS CN Workflow (Variance)	Show	Change Management	Y
KSC Revision Manager Change Activity	Show	Change Management	N
KSC Revision Manager CN Wflow	Show	Change Management	N
KSC SE Peer Review Workflow	Show	Change Management	Y
KSC Signature Required Change Activity	Show	Change Management	N
KSC Superseded CN Wflow	Show	Default	N
KSC System Safety & Reliability Analysis Impact Checklist WF	Show	Change Management	Y
KSC TRR Lite WF	Show	Change Management	N
KSC Variance Workflow KDP-KSC-P-1865 Rev A-2	Show	Change Management	Y
Main Translation Process	Show	Default	N
NE Problem Report - External Workflow	Show	Default	N
NE Problem Report - Internal Workflow	Show	Default	Y
NE Problem Report - RFI Workflow	Show	Change Management	N
Notification Route (KSC-PRJL)	Show	Default	N
Notify Only	Show	Default	N
Notify Process	Show	Default	N
OOTB_Problem Report Workflow	Show	Default	N
Package Approval Process	Show	Change Management	N
Package Content Routing Process	Show	Default	N
Package Delivery Process	Show	Change Management	N
Part Request Workflow	Show	Change Management	N
Peer Review Workflow	Show	Change Management	N
Post-Import Translation Process	Show	Default	N
Problem Report Workflow	Show	Default	N
Promotion Request Approval Process	Show	Default	N
Promotion Request Review Process	Show	Default	N
Quote Translation Process	Show	Default	N
Release Process	Show	Default	N
Review	Show	Default	N
Review Process	Show	Default	N
Revision Match Request	Show	Change Management	N
Submit			N
Submit PR			N
Two Level Approval Process			N
Variance Workflow			N

Glenn Research Center (GRC)

Source Windchill Instance (I.D./Name and Location)

- Source Instance ID / Name:
- Data Center / Location: Brookpark, Ohio
- Hosting/Virtualization (On-prem, VM type, hypervisor details): On Premise
- OS Version: Windows Server 2022

Windchill Version and Build

- Windchill Version/Build: 13.0.2.10
- Patch Level / Interim Fixes:

Modules and Scope

- Active Windchill Modules (PDMLink, Product Structure, BOM, etc.):
- Module-specific configurations:

Database and Infrastructure

- Database Platform (Oracle/SQL Server/Other): Oracle
- Database Version: 19C
- Database Size (TB):0.20
- Application Servers/Stack Details:
- Non-default Schemas or Special Configs: GRCPRODUCTION Schema

Data Volumes and Content

- Total Records/Object Counts: See Additional Notable Information
- Attachments Count and Total Size: See Additional Notable Information
- Number of Vaults & Sizes (TB):
 - 1 = 3.15 TB
 - 2 = 1.3 TB

Data Quality and Clean-up Needs

- Known Data Quality Issues (Duplicates, Orphans, etc.):
- Planned Cleansing/Normalization Steps (if any):

Customizations and Extensions

- Description/number of Custom Workflows/Rules/Events/Job Steps:
- Description/number of UI Customizations and CodeSnippets:
- Number of Custom subtypes and Attributes:

Integrations and Interfaces

- Interfaces and Data Exchange Cadence:
- CAD Integrations and Data Pipelines:

Org Structure and Governance

- Source Orgs Count:
- Proposed Org Mapping Strategy [Source Org → MCWC Org] (one-to-one or custom):

Cutover Readiness

- Proposed Cutover Windows and Downtime Tolerance:

Post-Migration Considerations

- Post-Migration Support Requirements:
- Post-Merge Compliance:

Additional Notes

- Any Known Constraints or Dependencies:

Additional Notable Information:

Select count(*) from EPMDOCUMENTMASTER – 601,750
 Select count(*) from EPMDOCUMENT – 1,502,270
 Select count(*) from WTDOCUMENTMASTER – 50,085
 Select count(*) from WTDOCUMENT – 78,786
 Select count(*) from WTPARTMASTER – 498,228
 Select count(*) from WTPART -- 1,108,453
 Select count(*) from WTCHANGEISSUEMASTER – 68
 Select count(*) from WTCHANGEISSUE – 68
 Select count(*) from WTCHANGEREQUEST2MASTER – 622
 Select count(*) from WTCHANGEREQUEST2 – 622
 Select count(*) from WTCHANGEORDER2MASTER – 635
 Select count(*) from WTCHANGEORDER2 – 639
 Select count(*) from WTCHANGEACTIVITY2MASTER – 704
 Select count(*) from WTCHANGEACTIVITY2 – 705
 Select count(*) from PROMOTIONNOTICE – 7,939
 Select count(*) from PROMOTIONSEED – 49,114
 Select count(*) from PROMOTIONTARGET – 128,504
 Select count(*) from WORKPACKAGEMASTER – 52
 Select count(*) from WORKPACKAGE – 55
 Select count(*) from DELIVERYRECORD – 53
 Application Data: 32,694,577
 Boolean Attribute Values: 2,246,118
 CAD Documents: 1,502,316
 Change Notices: 639
 Change Requests: 621
 Change Tasks: 705
 Container Teams: 758
 Documents: 78,752
 File Vault Folders: 500
 File Vault Items: 28,147,254
 File Vault Mounts: 501
 File Vaults: 1
 Groups: 12,811
 Libraries: 7

Managed Baselines: 388

Parts: 1,108,236

Problem Reports: 68

DRAFT

Langly Research Center (LaRC)

Source Windchill Instance (I.D./Name and Location)

- Source Instance ID / Name:
- Data Center / Location: Langley Research Center / Hampton, VA
- Hosting/Virtualization (On-prem, VM type, hypervisor details): On-Prem, VMware, vSphere Client version 8.0.3.00700 & Physical server Dell R750
- OS Version: Windows Server 2022 & RHEL 9.7

Windchill Version and Build

- Windchill Version/Build: 13.1 (upgrade in progress from 12.0.2.26)
- Patch Level / Interim Fixes: 13.1 (upgrade in progress from 12.0.2.26)

Modules and Scope

- Active Windchill Modules (PDMLink, Product Structure, BOM, etc.): PDMLink, Product Structure, BOM
- Creo View - Clients
 - windchill 12.0.2.26 REST services
 - windchill projectlink 12.0.2.7
- Creo View - Adapters 8.0.0.0-N63NhK
 - windchill 12.0.2.26 service pack
 - windchill pdmlink 12.0.2.7
 - windchill 12.0.2.16 info*engine
 - windchill 12.0.2.7 Services
 - windchill 12.0.2.7 common base
- Module-specific configurations: N/A

Database and Infrastructure

- Database Platform (Oracle/SQL Server/Other): Oracle
- Database Version: 19c (19.30)
- Database Size (TB): .27TB
 - Actual Data: 49 GB
 - Actual Index: 33 GB
 - Actual Blobs: 175 GB
 - WCAUDIT: 13 GB
- Application Servers/Stack Details: Linux 4.18.0-553.105.1.el8_10.x86_64
- Non-default Schemas or Special Configs: NONE

Data Volumes and Content

- Total Records/Object Counts: (See Below)
- Attachments Count and Total Size:
- Number of Vaults & Sizes (TB): 1 Vault (2.07TB)

Data Quality and Clean-up Needs

- Known Data Quality Issues (Duplicates, Orphans, etc.): N/A
- Planned Cleansing/Normalization Steps (if any): N/A

Customizations and Extensions

- Description/number of Custom Workflows/Rules/Events/Job Steps: 1 workflow (Electronic Drawing Release (created by PTC))
- Description/number of UI Customizations and CodeSnippets: N/A
- Number of Custom subtypes and Attributes: N/A

Integrations and Interfaces

- Interfaces and Data Exchange Cadence: N/A
- CAD Integrations and Data Pipelines: Creo, Zuken (in progress)

Org Structure and Governance

- Source Orgs Count: 1
- Proposed Org Mapping Strategy [Source Org → MCWC Org] (one-to-one or custom): 1

Cutover Readiness

- Proposed Cutover Windows and Downtime Tolerance: Cutover dictated by source, downtime limited to weekends (Friday 6pm-Monday 7am) and weekdays after hours (6pm-7am)

Post-Migration Considerations

- Post-Migration Support Requirements: Unknown
- Post-Merge Compliance: Unknown

Additional Notes

- Any Known Constraints or Dependencies: Currently undergoing a Windchill upgrade

Additional Notable Information:

Select count(*) from EPMDOCUMENTMASTER; 478751

Select count(*) from EPMDOCUMENT; 1437669

Select count(*) from WTDOCUMENTMASTER; 4485

Select count(*) from WTDOCUMENT; 4568

Select count(*) from WTPARTMASTER; 818

Select count(*) from WTPART; 1021

Select count(*) from WTCHANGEISSUEMASTER; 0

Select count(*) from WTCHANGEISSUE; 0

Select count(*) from WTCHANGEREQUEST2MASTER; 0

Select count(*) from WTCHANGEREQUEST2; 0

Select count(*) from WTCHANGEORDER2MASTER; 0

Select count(*) from WTCHANGEORDER2; 0

Select count(*) from WTCHANGEACTIVITY2MASTER; 0

Select count(*) from WTCHANGEACTIVITY2; 0

Select count(*) from PROMOTIONNOTICE; 4799

Select count(*) from PROMOTIONSEED; 9215

Select count(*) from PROMOTIONTARGET; 29660

Select count(*) from WORKPACKAGEMASTER; 0

Select count(*) from WORKPACKAGE; 0

Select count(*) from DELIVERYRECORD; 0

Application Data: 20259247

Boolean Attribute Values: 30

CAD Documents: 1437664

Change Notices: 0

Change Requests: 0

Change Tasks: 0

Container Teams: 213

Documents: 4574

File Vault Folders: 268

File Vault Items: 14189939

File Vault Mounts: 269

File Vaults: 2

Groups: 5356

Libraries: 5

Managed Baselines: 656

Parts: 1021

Problem Reports: 0